

ZIKAI XIONG

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Updated on September 21, 2026 (click [here](#) for the latest version)

ACADEMIC APPOINTMENTS

Northwestern University, Evanston, IL

July 2026 – present

ASSISTANT PROFESSOR

Department of Industrial Engineering and Management Sciences

Georgia Institute of Technology, Atlanta, GA

July 2025 – July 2026

POSTDOCTORAL FELLOW

School of Industrial and Systems Engineering

EDUCATION

Massachusetts Institute of Technology, Cambridge, MA

2020 – 2025

PH.D. IN OPERATIONS RESEARCH

Advisor: Prof. Robert M. Freund

Fudan University, Shanghai, China

2016 – 2020

B.S. IN MATHEMATICS AND APPLIED MATHEMATICS

RESEARCH INTERESTS

Twin interests in the mathematical foundations and computational practice of optimization, with particular interests in modern computational architectures and emerging interfaces between optimization and AI, and applications in operations research, data science, and AI.

PUBLICATIONS (alphabetical authorship marked with *)

PAPERS PUBLISHED OR UNDER REVIEW AT JOURNALS

- J1 Zikai Xiong. High-Probability Polynomial-Time Complexity of Restarted PDHG for Linear Programming. [arXiv] [pdf]
Mathematics of Operations Research (forthcoming)
- J2 Zikai Xiong. Accessible Complexity Bounds for Restarted PDHG on Linear Programs with a Unique Optimizer. [link] [pdf]
Mathematics of Operations Research, 2026
 - Winner, 2025 INFORMS Optimization Society Student Paper Prize
- J3 Zikai Xiong and Robert M. Freund. Computational Guarantees for Restarted PDHG for LP based on “Limiting Error Ratios” and LP Sharpness. [link] [pdf]
Mathematical Programming, 2026
 - Honorable Mention, 2024 INFORMS Optimization Society Student Paper Prize
- J4 *Dongdong Ge, Chengwenjian Wang, Zikai Xiong and Yinyu Ye. From an Interior Point to a Corner Point: Smart Crossover. [link] [pdf]
INFORMS Journal on Computing, 2025
- J5 Zikai Xiong and Robert M. Freund. Using Taylor-Approximated Gradients to Improve the Frank-Wolfe Method for Empirical Risk Minimization. [link] [pdf] [additional results]
SIAM Journal on Optimization, 2024
- J6 Zikai Xiong, Renjie Xu, Yanwei Xu and Yimin Wei. Low-rank Traffic Matrix Completion with Marginal Information. [link] [pdf]
Journal of Computational and Applied Mathematics, 2022

- J7 Zikai Xiong and Robert M. Freund. Symmetry-dependence in Rounding of a Convex Body. [arXiv] [pdf]
Under preparation for submission, 2026
- J8 *Katya Scheinberg and Zikai Xiong. Function-free Optimization via Comparison Oracles. [arXiv] [pdf]
Under review at *Mathematical Programming*, 2026
- J9 Zhenwei Lin, Zikai Xiong, Dongdong Ge and Yinyu Ye. A Practical GPU-Enhanced Matrix-Free Primal-Dual Method for Large-Scale Conic Programs. [arXiv] [pdf] [additional results] [software]
Under revision at *INFORMS Journal on Computing*, 2025
- J10 *Jiacheng Guo, Bo Jiang, Zikai Xiong and Nanxi Zhang. Two Sides of “Hidden City Ticketing”: Analysis of a Choice-Based Network Revenue Management Model. [link]
Under revision for resubmission at *Manufacturing & Service Operations Management*, 2025
- J11 Zikai Xiong and Robert M. Freund. The Role of Level-Set Geometry on the Performance of PDHG for Conic Linear Optimization. [arXiv] [pdf]
Split into two subsequent papers; one [link] is under review at *Mathematics of Operations Research*, 2026
- Finalist, 2024 George Nicholson Student Paper Competition
 - Honorable Mention, 2024 MIT ORC Best Student Paper Competition

PAPERS PUBLISHED AT CONFERENCES

- C1 Zikai Xiong, Niccolò Dalmaso, Shubham Sharma, Freddy Lecue, Daniele Magazzeni, Vamsi K. Potluru, Tucker Balch and Manuela Veloso. Fair Wasserstein Coresets. [link] [pdf]
Preliminary version appeared in NeurIPS 2023 Workshop SyntheticData4ML (Oral Presentation) Neural Information Processing Systems (NeurIPS), 2024
- C2 Zikai Xiong, Niccolò Dalmaso, Alan Mishler, Vamsi Potluru, Tucker Balch and Manuela Veloso. Fair-WASP: Fast and Optimal Fair Wasserstein Pre-processing. [link] [pdf]
AAAI Conference on Artificial Intelligence (AAAI), 2024
- C3 Zhengqi Gao, Fan-Keng Sun, Mingran Yang, Sucheng Ren, Zikai Xiong, et al. Learning from Multiple Annotator Noisy Labels via Sample-wise Label Fusion. [link] [pdf]
European Conference on Computer Vision (ECCV), 2022
- C4 *Dongdong Ge, Haoyue Wang, Zikai Xiong and Yinyu Ye. Interior-Point Methods Strike Back: Solving the Wasserstein Barycenter Problem. [link] [pdf]
Neural Information Processing Systems (NeurIPS), 2019

DISSERTATION

- D1 Zikai Xiong. New Theory and New Practical Methods for Solving Large-Scale Linear and Conic Optimization [link] [pdf], 2025
- Second Place, 2025 George B. Dantzig Dissertation Award

TECHNICAL REPORTS

- T1 Zhenwei Lin, Zikai Xiong, Dongdong Ge and Yinyu Ye. A Technical Note on the Implementation and Use of PDCS [arXiv] [pdf], 2026
- T2 Zikai Xiong and Robert M. Freund. On the Relation Between LP Sharpness and Limiting Error Ratio and Complexity Implications for Restarted PDHG. [arXiv] [pdf], 2023

PATENT APPLICATIONS

- P1 Niccolò Dalmaso, Zikai Xiong, Vamsi Krishna Potluru, Tucker Richard Balch and Manuela Veloso. Method and System for Generating Fair Synthetic Representative Data via Optimal Transport. [link]
U.S. Patent Application Publication No. US 2025/0181999 A1, pending, 2025.
- P2 Niccolò Dalmaso, Zikai Xiong, Alan Mishler, Vamsi Krishna Potluru, Tucker Richard Balch and Manuela Veloso. Method and System for Pre-processing Data for Algorithmic Fairness via Optimal Transport. [link]
U.S. Patent Application Publication No. US 2025/0173389 A1, pending, 2025.

SOFTWARE

- **PDCS**: a GPU-enhanced matrix-free primal-dual solver for large-scale conic programs, with JuMP and CVXPY interfaces. [link]
- **PDLP Complexity Bound**: a Julia toolkit that computes the accessible iteration bounds of restarted PDHG for LP. [link]
- **MAAIPM**: a MATLAB solver for exactly solving Wasserstein barycenter problems using interior-point methods. [link]

WORK EXPERIENCE

Georgia Institute of Technology, Postdoctoral Fellow, Atlanta, GA *July 2025 - July 2026*
Supervisor: Katya Scheinberg

- Function-free optimization with comparison oracles.

MIT Sloan School of Management, Research Assistant, Cambridge, MA *2020 - 2025*
Supervisor: Robert M. Freund

- Modern optimization methods for huge-scale operations research and machine learning problems.

J.P.Morgan Chase AI Research, Research Associate Intern, New York, NY *Summer 2023*
Managers: Niccolò Dalmaso, Vamsi Potluru and Manuela Veloso

- Algorithmic fairness in machine learning and large-scale optimization.

Shanghai University of Finance and Economics (SUFU), Research Assistant, Shanghai, China *2019 - 2020*
Supervisors: Yinyu Ye (Stanford), Dongdong Ge

- Optimization for operations research and machine learning.
- Developed and improved algorithms for the commercial solver COPT.

TEACHING

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Computing in Optimization and Statistics (15.S60) *Winter 2024*

- Instructor for the advanced optimization sessions of a computing course during MIT January Independent Activities Period. The duties include designing lectures, creating teaching materials, and giving lectures.

Introduction to Mathematical Programming (15.081) *Fall 2022 and Fall 2023*

- Head TA for MIT's doctoral introductory optimization course (for ORC and other MIT PhD programs).
- Teaching score: 6.8/7 (Fall 2023) and 6.6/7 (Fall 2022).
- Voted as the "Best TA" in the ORC Superlatives events (in 2023 and 2024).

The Analytics Edge (15.071) *Spring 2022*

- TA for MBA course in business analytics at Sloan School of Management.
- Teaching score: 6.4/7.

Online Course in Data Science and Business Analytics *Summer 2022*

- Designed course materials for a new online course in data science and business analytics.
- Supervisor: Robert M. Freund

GUEST LECTURE

- Large-Scale Linear Programming Using the Primal-Dual Hybrid Gradient Method
MIT, Cambridge, MA *November 2025*

TALKS

FUNCTION-FREE OPTIMIZATION VIA COMPARISON ORACLES

- Modeling and Optimization: Theory and Applications (MOPTA), Lehigh University, PA *August 2026*
- Foundations of Computational Mathematics (poster), University of Vienna, Austria *July 2026*

- Third Derivative-Free Optimization Symposium, Université de Montréal *June 2026*
- SIAM Conference on Optimization (OP26), Edinburgh, United Kingdom *June 2026*
- INFORMS Optimization Society Conference, Atlanta, GA *March 2026*

PDCS: A PRIMAL-DUAL LARGE-SCALE CONIC PROGRAMMING SOLVER WITH GPU ENHANCEMENTS

- INFORMS Annual Meeting, Atlanta, GA, *October 2025*
- cuOPT Team, NVIDIA, virtual *July 2025*
- Computational Robotics Group Seminar, Harvard University, Cambridge, MA *May 2025*

NEW THEORY AND NEW PRACTICAL METHODS FOR SOLVING LARGE-SCALE LINEAR AND CONIC OPTIMIZATION

- INFORMS Annual Meeting (Dantzig Dissertation Competition Session), Atlanta, GA *October 2025*
- PhD Thesis Defense, MIT, Cambridge, MA *May 2025*

THEORETICAL FOUNDATIONS AND PRACTICAL IMPROVEMENTS FOR LARGE-SCALE LINEAR PROGRAMMING

- Data Science Lab Seminar, MIT, Cambridge, MA *April 2025*
- Georgia Institute of Technology, Atlanta, GA *April 2025*
- Columbia University, New York, NY *February 2025*
- University of Chicago, Chicago, IL *February 2025*
- Cornell University, Ithaca, NY *January 2025*
- Northwestern University, Evanston, IL *January 2025*
- University of British Columbia, Vancouver, BC, Canada *January 2025*
- NC State University, Raleigh, NC *January 2025*
- Chinese University of Hong Kong (CUHK), Hong Kong *January 2025*
- Hong Kong University of Science and Technology (HKUST), Hong Kong *December 2024*

LEVEL-SET GEOMETRY AND IMPROVING THE PERFORMANCE OF PDHG FOR LINEAR OPTIMIZATION

- INFORMS Nicholson Student Paper Competition, Seattle, WA *October 2024*
- INFORMS Annual Meeting, Seattle, WA *October 2024*
- Cornell ORIE Young Researchers Workshop, Ithaca, NY *October 2024*
- OM Seminar, MIT, Cambridge, MA *September 2024*

THEORETICAL COMPLEXITY OF PDHG FOR LINEAR PROGRAMS WITH UNIQUE OPTIMA

- INFORMS Annual Meeting (Optimization Society Award Session), Atlanta, GA *October 2025*
- ARC Colloquium, Georgia Tech, Atlanta, GA *September 2025*
- Modeling and Optimization: Theory and Applications (MOPTA), Lehigh University, PA *August 2024*
- International Symposium on Mathematical Programming (ISMP), Montréal, Canada *July 2024*
- Conference in honor of Yinyu Ye's retirement, Shanghai, China *July 2024*

IMPROVING THE GEOMETRY OF (CONIC) LINEAR OPTIMIZATION PROBLEMS FOR THE PRIMAL-DUAL HYBRID GRADIENT METHOD

- Modeling and Optimization: Theory and Applications (MOPTA), Lehigh University, PA *August 2024*
- INFORMS Optimization Society Conference, Houston, TX *March 2024*
- LIDS Conference, MIT, Cambridge, MA *February 2024*
- INFORMS Annual Meeting, Phoenix, AZ *October 2023*
- ORC Student Seminar, MIT, Cambridge, MA *October 2023*
- Shanghai University of Finance and Economics, Shanghai, China *September 2023*
- Workshop on Modern Continuous Optimization, Cambridge, MA *August 2023*

GEOMETRIC CONDITION MEASURES FOR THE PRIMAL-DUAL HYBRID GRADIENT METHOD FOR LINEAR PROGRAMMING

- SIAM Conference on Optimization (SIOPT), Seattle, WA *June 2023*
- ORC Student Seminar, MIT, Cambridge, MA *March 2023*

USING TAYLOR-APPROXIMATED GRADIENTS TO IMPROVE THE FRANK-WOLFE METHOD FOR EMPIRICAL RISK MINIMIZATION

- INFORMS Annual Meeting, Indianapolis, IN *October 2022*
- ORC Student Seminar, MIT, Cambridge, MA *October 2022*
- International Conference on Continuous Optimization (ICCOPT), Lehigh University, Bethlehem, PA *July 2022*

FROM AN INTERIOR POINT TO A CORNER POINT: SMART CROSSOVER

- INFORMS Annual Meeting, Indianapolis, IN

October 2022

INTERIOR-POINT METHODS STRIKE BACK: SOLVING THE WASSERSTEIN BARYCENTER PROBLEM

- INFORMS Annual Meeting, Seattle, WA
- “Heart of the Machine” Seminar, Shanghai, China

October 2019

October 2019

COMPUTING THE WASSERSTEIN BARYCENTER EFFICIENTLY: A STRUCTURED LINEAR SYSTEM AND CUSTOMIZED ALGORITHMS

- Seminar of New Advances in Theory and Application of ADMM, Shanghai, China

March 2019

SERVICE

REFeree for

- Operations Research
- Mathematics of Operations Research
- SIAM Journal on Optimization
- Mathematical Programming
- INFORMS Journal on Computing
- Journal of Optimization Theory and Applications
- Tutorials in Operations Research
- Optimization Letters
- Journal of Machine Learning Research
- AAAI Conference on Artificial Intelligence (AAAI)
- International Conference on Machine Learning (ICML)
- Conference on Neural Information Processing Systems (NeurIPS)

SESSION (CO-)CHAIR for

- INFORMS Annual Meeting, San Francisco, CA
- Modeling and Optimization: Theory and Applications (MOPTA), Bethlehem, PA
- INFORMS Optimization Society Conference, Atlanta, GA
- INFORMS Annual Meeting, Seattle, WA
- International Symposium on Mathematical Programming (ISMP), Montréal, Canada
- INFORMS Optimization Society Conference, Houston, TX
- INFORMS Annual Meeting, Indianapolis, IN

October 2026

August 2026

March 2026

October 2024

July 2024

March 2024

October 2022

AWARDS AND HONORS

- NSF Travel Award for Young Researchers, Foundations of Computational Mathematics (FoCM) *2026*
- **Winner, INFORMS Optimization Society Student Paper Prize** *2025*
- **Second Place, George B. Dantzig Dissertation Award** *2025*
- **Finalist, George Nicholson Student Paper Competition** *2024*
- **Honorable Mention, INFORMS Optimization Society Student Paper Prize** *2024*
- **Honorable Mention, MIT ORC Best Student Paper Competition** *2024*
- INFORMS Optimization Society Conference Travel Award *2024*
- MIT Graduate Student Council Travel Grant *2022*
- First Prize, MIT ORC Common Experience Deep Learning Challenge *2021*
- SIAM Travel Award *2021*
- Star Graduate Award, Fudan University *2020*
- Outstanding Graduate of Shanghai City *2020*
- Top Outstanding Undergraduate Student Award, Fudan University *2019*

OTHER

Programming languages: Julia, Python, MATLAB, R

Personal interests: Hiking, skiing, traveling, and cooking Sichuan cuisine.

REFERENCES

KATYA SCHEINBERG, Georgia Institute of Technology (Postdoctoral Supervisor)
Coca-Cola Foundation Chair and Professor
H. Milton Stewart School of Industrial and Systems Engineering
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ROBERT M FREUND, MIT (Ph.D. Advisor)
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